Heat-Aware Pedestrian Routing with Raster-Derived Exposure and Hard Detour Budgets: A Washington, DC Pilot

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Abstract

Heat-aware walking routes must balance modeled thermal exposure against extra distance. This pilot samples Universal Thermal Climate Index (UTCI) and shadow rasters on a Washington, DC walking network and compares four environmental objectives with shortest distance at 5–30% detour allowances. One modeled hour and 500 origin–destination pairs produce 10,000 attempts; 6,093 of 8,000 environmental attempts return an accepted route. A reanalysis of the original records compares all five methods on 1,386 common successful pair–budget records. Mean reductions in length-weighted UTCI exceedance above 26°C range from 0.576 K for the composite objective to 0.701 K for the exceedance-focused objective. A separate, limited weight-tuning run changes 25–30% of common accepted paths but improves this metric by only 0.044–0.050 K relative to its default. All 1,907 main-run failures log an expansion-limit error. These results characterize a capped routing implementation under one modeled snapshot. They do not establish transfer across cities, physiological benefit, or robustness to sampling and threshold choices. Code, original route records, and reanalysis scripts accompany the paper.

Keywords

pedestrian routing, UTCI, shade, detour constraints, computational benchmark

1 Introduction

Two walking routes of similar length can have different shade and thermal conditions. A route that avoids exposed streets may reduce modeled heat exposure, but a long detour also adds walking. An explicit distance allowance makes this tradeoff understandable: a traveler can ask for a cooler route no more than 10% longer than the shortest path.

This paper examines that formulation in a Washington, DC pilot. It attaches modeled UTCI and direct-shadow rasters to a pedestrian graph, then compares objectives that emphasize UTCI, shade, or a composite of environmental features. The contribution is a computational benchmark of raster-to-network integration and bounded search. Hard detour constraints, multi-label search, and environmentally enriched graphs are established ideas; the paper does not introduce a new routing algorithm.

The experiment uses 500 ordered origin–destination (OD) pairs, four detour allowances, and five methods. The original method-level summaries used different accepted subsets, making their means unsuitable for ranking methods. This revision recovers the raw records and makes the comparison on a common successful subset. It also examines logged failures, checks baseline edge sequences, and reanalyzes an existing weight-tuning experiment. These additions answer questions about the historical run; they are not a rerun with the corrected implementation.

The distinction matters because the implementation audit found an internal distance ceiling that could exceed the final acceptance ceiling, a parallel-edge fallback issue, and a difference between two UTCI exceedance calculations. Section 3.4 describes these behaviors. The corrected code preserves the original-network ceiling and sampled exceedance, but the numerical results here remain those recorded by the historical pipeline.

2 Related Work

Shade-aware routing predates this study. Vartholomaios derives least-cost paths from SOLWEIG shadows [21]; Da et al. use image-derived shade [5]; and Kolaxidis et al. incorporate solar exposure into pedestrian routing [13]. CoolWalks studies sunlit-distance penalties and the difference between centerline and detailed pedestrian networks [23]. Yang et al. update shade costs at expected arrival times [24], whereas this pilot holds the environmental field fixed.

Other work uses thermal indices. Colaninno et al. combine sidewalk mobility and microclimate modeling [4]. Patel et al. rank candidate routes using SOLWEIG-GPU UTCI [19]. CoolPaths uses a related hard-detour formulation with length-weighted Physiological Equivalent Temperature (PET) [22]. Our use of UTCI, direct shadow, and additional engineering proxies changes the inputs and objectives, not the underlying constrained-routing problem. More generally, Skoumas et al. enrich route computation with contextual information about places [20]; environmental attributes are another application of that idea.

UTCI converts meteorological and radiative conditions into an equivalent temperature under standardized reference assumptions [3]. SOLWEIG models urban radiation and mean radiant temperature [15]. Evidence that heat influences route choice [1] and that pedestrians seek shade [14] motivates these routing preferences. It does not calibrate this paper's weights or validate its route-level outcomes for an individual pedestrian.

3 Routing Pipeline

3.1 Graph and distance budget

Let G_0 be the original directed walking multigraph. For a route P, let $L(P) = \sum_{e \in P} l_e$, where l_e is edge length in metres. For an OD pair, define the original-network shortest distance L_0^* and the final ceiling

$$L_0^* = \min_{P \subset G_0} L(P), \qquad D_0 = (1 + \delta) L_0^*. \tag{1}$$

The minimization is over directed paths connecting that pair. The allowance $\delta \in \{0.05, 0.10, 0.20, 0.30\}$ limits proportional extra distance.

Figure 1 shows the four pipeline stages. To reduce feature-extraction work, a node v is retained when

$$d_{G_0}(o, v) + d_{G_0}(v, d) \le D_0 + 50\,\mathrm{m}, \tag{2}$$

where d_G denotes shortest-path distance on graph G. The pipeline then assigns raster-backed features and removes edges lacking valid UTCI samples. The resulting feature graph is G_f. Network processing uses OpenStreetMap, OSMnx, and NetworkX [2, 11, 18].

3.2 Sampling and reported exposure

Each source segment in an edge geometry is subdivided to a maximum spacing of 20 m, retaining endpoints. Coordinates are transformed into each raster's coordinate reference system before sampling. Non-finite and NoData samples are omitted. Edges without UTCI samples are removed; edges without direct-shadow samples use the surrogate in Appendix A.

For UTCI samples u_{ei} on edge e, define

$$\bar{u}_e = \frac{1}{n_e^u} \sum_i u_{ei}, \qquad u_e^{\max} = \max_i u_{ei}, \tag{3}$$

$$h_e = \frac{1}{n_e^u} \sum_i \max(0, u_{ei} - \tau), \quad \tau = 26^\circ\mathrm{C}. \tag{4}$$

For valid shadow samples b_{ej}, the direct-shade fraction is $s_e^{\mathrm{dir}} = (n_e^b)^{-1} \sum_j \mathbf{1}[b_{ej} \le 0.5]$; the raster convention is zero for shade and one for sun. Except for hybrid azimuth, each method uses this value when present and the surrogate s_e^{az} otherwise. The hybrid-azimuth method uses s_e^{az} everywhere.

The 26°C reference is the conventional UTCI boundary between no thermal stress and moderate heat stress [10]. Its use here is an engineering reference, not a personalized safety threshold. The 20 m interval reduces sampling work on nominal 1 m rasters. It can miss short shaded segments; no convergence study establishes its accuracy.

Reported route quantities are length-weighted edge summaries:

$$H(P) = \frac{\sum_e l_e h_e}{L(P)}, \qquad \bar{U}(P) = \frac{\sum_e l_e \bar{u}_e}{L(P)}, \tag{5}$$

$$S(P) = \frac{\sum_e l_e s_e}{L(P)}, \qquad U^{\max}(P) = \max_e u_e^{\max}. \tag{6}$$

H is mean positive UTCI exceedance, expressed in kelvins; $\bar{U}$ and $U^{\max}$ are UTCI values in degrees Celsius. S is a shade fraction. Because sampling restarts at segment boundaries, these quantities are numerical proxies rather than uniform line integrals. In particular, H is not a time-integrated dose: a longer route with lower H can still entail more total exposure.

3.3 Objectives and method variants

The composite cost is our engineering construction. It combines distance and time with thermal, scenic, traffic, and slope penalties; Appendix B specifies the terms and normalizations. All are non-negative. The thermal term is

$$\phi_e = \frac{[\bar{u}_e - \tau]_+}{8\,\mathrm{K}} + \frac{[u_e^{\max} - \tau]_+}{10\,\mathrm{K}} + \frac{h_e}{6\,\mathrm{K}} + 0.45(1 - s_e), \tag{7}$$

where $[x]_+ = \max(0, x)$. Its correlated UTCI and shade components can count related radiative information more than once. Neither its coefficients nor the contextual proxies are fitted to route-choice or physiological observations.

Table 1: Objectives used in the historical experiment. A_e is the distance–time term in Appendix B; all environmental penalties also use its edge-length scale.

Method	Selection objective
Shortest	Dijkstra search minimizing $L(P)$.
Hybrid azimuth	A_e plus twice the thermal term, using s_e^{az} on all edges.
Direct shadow	A_e plus twice the shade-only part of ϕ_e.
UTCI exceedance	A_e plus twice the $h_e/(6\,\mathrm{K})$ part of ϕ_e.
Composite	Full thermal, scenic, traffic, and slope cost.

Table 1 treats the environmental methods as objective variants of one implementation. Distance and time remain active in all four. The focused methods are therefore not pure single-feature optimizations. For reporting, selected routes are evaluated with full features; the hybrid shade outcome remains a surrogate and is not comparable with direct shade.

3.4 Search, acceptance, and implementation audit

In the historical formulation, the internal ceiling is $D_f = (1 + \delta)L_f^*$, where L_f^* is shortest distance on G_f. Removing edges cannot shorten a shortest path, so $L_f^* \ge L_0^*$ and $D_f \ge D_0$. The internal search can therefore admit a candidate that fails the final original-network check.

The solver first computes an unconstrained minimum-cost path and returns it if it meets D_f. Otherwise it explores cost–distance labels. At a node, a label dominates another if it has no greater length and no greater cost. Labels are also pruned when accumulated length plus shortest remaining distance exceeds D_f. Paths may not repeat nodes.

The dominance registry retains at most eight labels per node, ordered by cost then length. A newly queued label can survive even when omitted from that registry, so this is not a bound on the queue size. The label loop stops after 20,000 queue removals or 1.5 s, with timing beginning after the initial cost-path calculation. These caps remove an optimality guarantee and do not bound end-to-end latency.

The inspected pre-revision fallback obtains a length-shortest node sequence but chooses the lowest-cost parallel edge between consecutive nodes. That can produce a longer edge sequence than the shortest one. It also projects h_e to $[\bar{u}_e - \tau]_+$ during objective selection; in general this differs from mean positive sampled exceedance. The reporting metric above retains the sampled definition. The historical execution commit was not recorded, so the audit cannot establish how much these discrepancies affected that run.

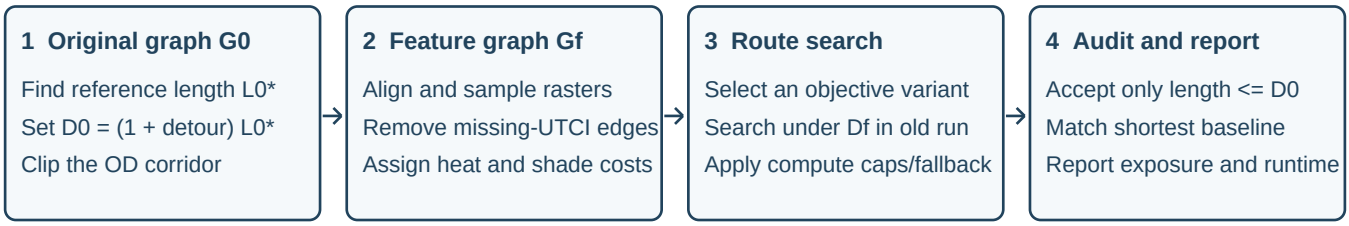


Figure 1: Historical routing pipeline. G_0 supplies the final budget D_0; the raster-covered graph G_f supplies the historical internal budget D_f. A returned path must still pass the D_0 check.

Table 2: Input layers. Different source years and street-centerline sampling limit spatial interpretation.

Layer	Source and role
Walking network	OpenStreetMap via Geofabrik/OSMnx; directed centerlines, not verified sidewalks [2, 18].
UTCI and shadow	SOLWEIG-GPU modeled snapshot; thermal and direct-shade features [12].
Terrain/buildings	NOAA-hosted DC OCTO 2022 LiDAR and DEM; grade and height proxies [16, 17].
Canopy/land cover	LiDAR canopy heights and DC 2020 layers; greenery, water, and surrogate shade [6, 7].
Traffic	DDOT 2023 AADT lines; normalized traffic proxy, not sound measurements [8].

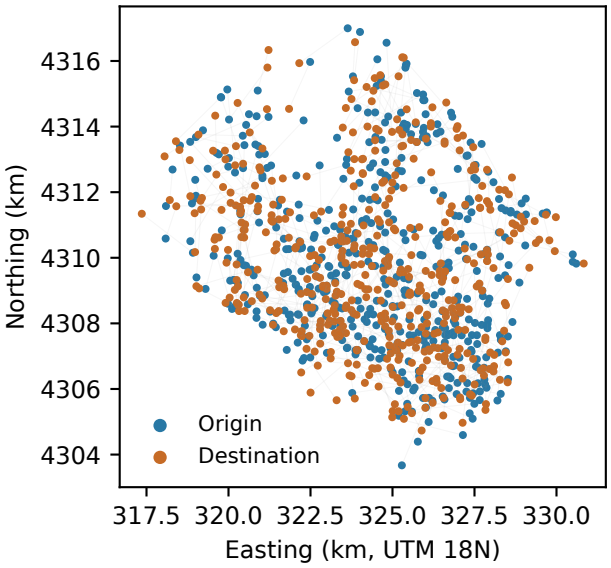


Figure 2: The original 500 OD pairs in metric coordinates. Endpoints are constrained by raster coverage. Gray connectors are not routes; eligibility uses network-edge lengths.

The corrected implementation propagates D_0 throughout search, uses shortest parallel edges for shortest-route fallback, and preserves sampled exceedance. It also records path source, cap termination, and rejection type. These fixes require a fresh real-data experiment. The present analysis keeps the historical outcomes separate. An accepted historical route passes the final distance check, but its unmarked source may be the initial cost path, label search, or fallback.

4 Data and Analysis

4.1 Study domain and inputs

The study represents Washington, DC on June 19, 2026 at an intended civil time of 14:00. Walking is relevant in this setting: Sustainable DC 2.0 reports a 13.3% walking commute share in 2016 [9]. This historical district-wide statistic does not make the sampled OD pairs representative of actual trips.

Table 2 lists the input sources. SOLWEIG-GPU [12] generated the nominal 1 m UTCI and shadow fields from urban-surface inputs and hourly Open-Meteo forcing [25]. The fields were not independently validated here. The retained provenance does not pin the forecast retrieval time, automatically selected forecast model, or environmental-model release. The routing timestamp has no UTC offset; exact recovery of the hybrid solar-position calculation would require the historical host offset.

4.2 OD sample and experimental design

The generator uses the largest weakly connected graph component, then retains endpoints with finite UTCI and shadow raster values. With seed 42, it samples 500 distinct ordered node pairs whose shortest network distances lie between 400 and 3,000 m. The recovered distances range from 412.9 to 2,998.9 m. Figure 2 redraws the original coordinates in UTM zone 18N (EPSG:32618); its connectors indicate pairs, not walking paths or the distance used for eligibility.

Each pair is evaluated with five methods at four allowances, giving 10,000 attempts. The 2,000 shortest-distance records repeat 500 pairs; recovered edge sequences confirm an identical baseline path across all four allowances for every pair. Environmental routes can change with allowance-specific corridors, features, and search. Records from the same OD pair are not independent observations.

4.3 Matched comparison and uncertainty

Let $P^0_{i\delta}$ be the matched shortest route and $P^m_{i\delta}$ the route returned by method m. Positive improvements are

$$\Delta H_{i\delta m} = H(P^0_{i\delta}) - H(P^m_{i\delta}), \tag{8}$$

with analogous $\Delta\bar{U}$; shade change reverses the order. Extra distance is $100(L(P^m_{i\delta})/L^*_0 - 1)$.

At each allowance, the primary comparison uses the intersection $\mathcal{C}_\delta$ of successful records for all five methods. Matching includes study, configuration hash, run identifier, raster timestamp, OD identifier, and allowance; duplicates are rejected. The common set contains 389, 318, 309, and 370 OD pairs at 5%, 10%, 20%, and 30%, respectively. Pooled means weight the resulting 1,386 pair–budget records equally and cover 459 distinct OD pairs. They describe this selected subset, not every attempted trip.

For a paired method contrast, we compute the mean difference in recorded H on the same common records. Percentile intervals use 2,000 bootstrap resamples of whole OD pairs, retaining each pair's available allowances, with seed 42. These exploratory 95% intervals are not adjusted for multiple comparisons. They summarize variability within this one modeled snapshot and do not account for microclimate error or selection of the successful subset.

4.4 Weight-tuning reanalysis

The archive also contains 800 attempts from the first 100 serialized OD pairs at 10% and 20% allowances. Four composite configurations jointly change several weights (Table 3). All other recorded study identifiers match the main run. The four variants share 142 successful pair–budget records from 81 pairs; all recorded successful paths meet the original distance ceiling. We compare variants on this intersection and score their recorded H under the same threshold and field. Path change means a different ordered edge sequence.

This is a limited joint-weight exploration, not an independent validation set or a one-factor sensitivity study. The archive does not establish whether the alternatives were selected before inspecting results. Threshold, spacing, and corrected-code sensitivity runs remain outstanding. We exclude the separate September uncertainty experiments because they use a different OD cohort, network, fields, threshold, and routing design.

5 Results

5.1 Acceptance and failure records

Table 4 shows counts before filtering failures. All shortest-distance attempts succeed. Environmental methods return 6,093 accepted routes from 8,000 attempts (76.2%). The composite has the highest count, but roughly one quarter of environmental attempts are rejected.

All 1,907 failed records contain the same message: the search exceeded 20,000 expansions before finding a route within the detour budget. This identifies the logged termination category, not a proof of geographic infeasibility or the full causal role of coverage and budget handling. Expansion counters and fallback-source flags are blank in the historical records.

Every original-graph OD pair has its own shortest path within D_0. Consequently, failure to return an environmental route does not mean that the neighborhood has no walkable route or no cooler alternative. Feature removal can eliminate the baseline, and the capped procedure may stop before finding another feasible path. This acceptance level is a material implementation limitation, not evidence that one quarter of trips are geographically infeasible.

5.2 Exposure and distance on the common set

Table 5 replaces the original comparison across unequal accepted subsets. All four variants reduce mean recorded H relative to shortest distance on the common set. The UTCI-exceedance and direct-shadow objectives give similar reductions, while the composite adds less distance on average.

On paired records, the exceedance-focused objective has 0.125 K lower mean H than the composite (OD-cluster interval 0.108–0.142 K). The difference between exceedance-focused and direct-shadow objectives is 0.008 K, with an interval spanning zero (−0.002 to 0.018 K). The latter comparison does not resolve a consistent separation between those objectives. These are comparisons of recorded exposure outcomes, not evidence that one objective dominates across distance, shade, or other preferences.

Positive averages also conceal adverse outcomes. In the full method-specific accepted sets, the 10th-percentile ΔH is negative for every environmental variant. Median reduction in maximum sampled UTCI is zero for each variant. Thus lower mean exposure does not imply improvement on every route or removal of its hottest segment. The hybrid's apparent shade improvement is especially unsuitable for ranking because its reporting field differs; direct-shadow methods can also substitute the surrogate on edges with missing shadow samples, with no historical substitution count.

5.3 Detour allowances

For the composite's method-specific accepted records, mean ΔH rises from 0.411 K at 5% allowance to 0.756 K at 30%, while mean extra distance rises from 3.37% to 14.50%. Acceptance is non-monotonic: 401, 349, 376, and 428 of 500 attempts succeed at the four allowances. These changing subsets do not identify a within-pair dose–response curve.

On a fixed graph, the set of paths meeting the final distance inequality grows with the allowance. The historical experiment nevertheless rebuilds its corridor, uses an internal feature-graph ceiling, and applies search caps. Its non-monotonic acceptance therefore reflects the implementation rather than a failure of that nesting property. A corrected rerun and cap sweep are needed to distinguish the contributions of those stages.

5.4 Sensitivity to joint weight changes

Table 6 summarizes the common accepted tuning records. The alternative weights change 36–42 of 142 paths (25.4–29.6%). Compared with the default, they lower mean H by 0.044–0.050 K, while mean added distance increases by about 0.25–0.28 percentage points. OD-cluster intervals for the additional reductions are 0.017–0.085 K, 0.020–0.088 K, and 0.023–0.091 K, respectively.

The tested configurations are close in this summary metric, but path choice is not invariant to the weights. Because several coefficients change together and the sample is small and selected, these

Table 3: Historical composite weight configurations. Columns multiply thermal, scenic, traffic, slope, time, and distance terms; shade multiplies the $0.45(1 - s_e)$ term inside ϕ_e. The uncertainty coefficient is 0.15 throughout, but its input is zero.

Configuration	Thermal	Shade	Scenic	Traffic	Slope	Time	Distance
Default	1.00	1.00	0.45	0.45	0.35	0.20	0.15
Heat-heavy	1.80	1.00	0.25	0.25	0.20	0.12	0.08
Shade-heavy	1.40	2.20	0.25	0.25	0.20	0.12	0.08
Low distance penalty	2.00	1.50	0.20	0.20	0.15	0.05	0.03

Table 4: Main-run outcomes. Every method has 2,000 attempts: 500 pairs at four budgets.

Method	Accepted	Rejected	Accepted (%)
Shortest	2,000	0	100.0
Hybrid azimuth	1,522	478	76.1
Direct shadow	1,538	462	76.9
UTCI exceedance	1,479	521	74.0
Composite	1,554	446	77.7

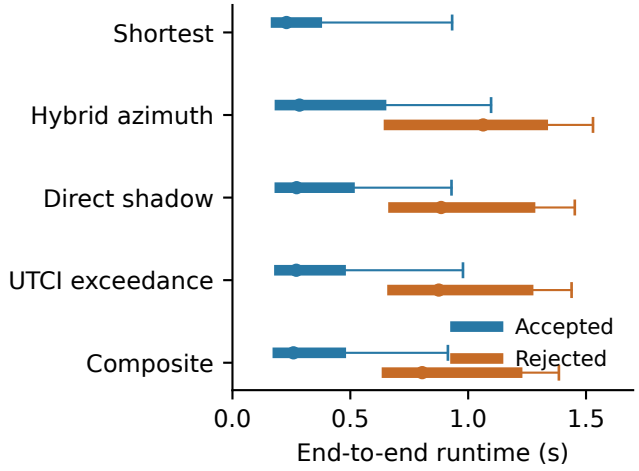


Figure 3: Observed end-to-end runtime. Thick bars show the 25th–75th percentiles, circles the medians, and thin lines extend to the 90th percentiles. Rejected attempts are shown separately. No density is inferred from these summaries.

results cannot identify an optimal setting or establish general robustness. They also do not address the separate choices of 26°C and 20 m.

5.5 Observed runtime

Figure 3 uses observed runtimes for accepted and rejected attempts. Accepted medians range from 0.229 to 0.284 s and 90th percentiles from 0.914 to 1.097 s. Rejected environmental attempts have medians of 0.805–1.064 s and 90th percentiles of 1.385–1.530 s. Their omission from successful-only summaries would understate unsuccessful-request cost.

Accepted means exceed medians, and upper quantiles extend much farther from the median than lower quartiles, indicating long upper tails. Timings include corridor and feature-graph work as well as routing. Direct-shadow methods share a persistent feature cache; hybrid azimuth uses another cache. Mixed cache states, execution order, and the unpinned hardware environment prevent a controlled solver-speed comparison. The 1.5 s label-loop cap is not an end-to-end latency guarantee.

6 Discussion and Limitations

The common-set reanalysis supports a narrow result: under this modeled field, focused thermal objectives reduce recorded mean UTCI exceedance more than the default composite on OD–budget records where every method succeeds. The composite's mean distance overhead is smaller. The analysis does not establish a universally best route, because the objectives encode different preferences and the successful subset excludes difficult attempts. It also does not establish that composite thermal reductions purchase better scenery, quieter streets, or easier grades; those interpretations would require a dedicated comparison of the corresponding outcomes.

The failure log strengthens the diagnosis of the acceptance problem. The recorded failure category is computational, and a public-facing router should distinguish a capped search from a proof that no feasible path exists. It should also identify shortest-route fallback explicitly. Exact or high-cap comparisons on smaller graphs would quantify approximation error; the current benchmark has no optimality-gap measurement.

The spatial and temporal scope remains limited to one modeled DC hour. The experiment cannot establish transfer across cities, seasons, neighborhoods, or times of day. A fixed raster also ignores changes during a walk. The original sampling targets covered graph nodes, not observed pedestrian demand, so its outcomes should not be interpreted as population benefits.

Fine raster resolution does not remove representation error. Street centerlines may lie in sun while a sidewalk is shaded, or the reverse. Source layers have different vintages, the traffic join uses a nearest-midpoint proxy in geographic coordinates without a maximum distance, and nominal canopy does not guarantee shade at the pedestrian's position. The hybrid shade value mixes geometry and cool UTCI samples; it is not a substitute for independently validated direct shadow.

The composite is uncalibrated. Thermal terms overlap, scenic and traffic scores are engineering proxies, and the uncertainty input is zero in this experiment. The limited tuning analysis adds evidence about three joint weight alternatives, but not about individual coefficients, threshold choice, or sampling convergence. Those questions require rerouting fixed OD pairs with controlled

Table 5: Comparison on the same 1,386 successful pair–budget records per method (459 OD pairs). ΔH and $\Delta \bar{U}$ are improvements over the matched shortest route. These are conditional pooled means, not results for all 500 pairs. The hybrid shade value uses a different surrogate and is marked separately.

Method	Mean ΔH (K)	Median ΔH (K)	Mean $\Delta \bar{U}$ (K)	Mean ΔS	Mean extra distance (%)
Shortest	0.000	0.000	0.000	0.000	0.00
Hybrid azimuth	0.639	0.505	0.646	0.477*	8.70
Direct shadow	0.693	0.557	0.699	0.107	9.30
UTCI exceedance	0.701	0.573	0.708	0.104	9.27
Composite	0.576	0.427	0.582	0.086	8.32

Table 6: Weight tuning on 142 common successful pair–budget records. Acceptance counts are out of 200 attempts per configuration.

Configuration	Accepted	ΔH (K)	Extra (%)	Changed
Default	145	0.705	9.10	—
Heat-heavy	146	0.749	9.36	36
Shade-heavy	146	0.752	9.37	36
Low distance	144	0.755	9.38	42

perturbations and scoring every selected path on a common reference field. The revised code supplies that workflow; no missing run is represented as completed here.

Finally, modeled exposure, route-choice preferences, and physiological outcomes are distinct validation targets. This pilot has no field transects, participant study, or health measurements. A reduction in length-weighted H does not establish reduced total dose, comfort, safety, or clinical benefit. Such claims would require separate environmental and human validation. A useful deployment would present distance, expected duration, shade, and modeled exposure together, with traveler-specific constraints and an accessible pedestrian network.

7 Conclusion

This pilot connects UTCI and shadow rasters to a walking network under explicit detour allowances. Reanalysis of the original route records resolves the unequal-subset comparison and identifies the logged search-limit failures. On 1,386 common successful OD–budget records, mean exceedance reductions are 0.576–0.701 K with 8.32–9.30% extra distance. A small joint-weight experiment changes a quarter to a third of its common paths while modestly changing the exposure summary. The remaining priorities are a corrected real-data rerun, threshold and sampling checks, and evaluation across additional snapshots. The present evidence supports a reproducible implementation case study with a defined scope.

Code and Data Availability

Code is at https://github.com/kameronlee/Routing. The revision branch `codex/geohealth-3466-revisions` contains the archived OD pairs, compressed main and tuning records, checksums, and reanalysis outputs under `paper_assets/review3466`. The analysis command is `experiments/reanalyze_geohealth3466.py`; it does not rerun routing. The historical records identify configuration `5334dd51b5f7`, but not an immutable execution commit or complete package/hardware environment. The corrected code was tested at `e1701ad`; its outputs must be distinguished from the archived run. Large graph and raster entries use Git LFS; pointers are not usable datasets, and regeneration alone would not recover the original frozen inputs.

References

[1] R. Basu, N. Colaninno, A. Alhassan, and A. Sevtsuk. 2024. Hot and bothered: Exploring the effect of heat on pedestrian route choice behavior and accessibility. *Cities* 155 (2024), 105435. doi:10.1016/j.cities.2024.105435

[2] G. Boeing. 2017. OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems* 65 (2017), 126–139. doi:10.1016/j.compenvurbsys.2017.05.004

[3] P. Bröde, D. Fiala, K. Błażejczyk, I. Holmér, G. Jendritzky, B. Kampmann, B. Tinz, and G. Havenith. 2012. Deriving the operational procedure for the Universal Thermal Climate Index (UTCI). *International Journal of Biometeorology* 56, 3 (2012), 481–494. doi:10.1007/s00484-011-0454-1

[4] N. Colaninno, R. Basu, M. Hosseini, A. Alhassan, L. Liu, and A. Sevtsuk. 2025. A sidewalk-level urban heat risk assessment framework using pedestrian mobility and urban microclimate modeling. *Environment and Planning B: Urban Analytics and City Science* 52, 5 (2025), 1071–1090. doi:10.1177/23998083241280746

[5] L. Da, R. Chhibba, R. Jaiswal, A. Middel, and H. Wei. 2024. Shaded route planning using active segmentation and identification of satellite images. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 5205–5209. doi:10.1145/3627673.3679234

[6] DC Office of the Chief Technology Officer. 2020. Generalized Land Cover 2020. ArcGIS image service, Open Data DC. Created by PlanIT Geo for the District Department of Transportation Urban Forestry Division; accessed 17 July 2026. https://imagery.dcgis.dc.gov/dcgis/rest/services/UFD/Generalized_Land_Cover_2020/ImageServer

[7] DC Office of the Chief Technology Officer. 2020. Urban Tree Canopy 2020. ArcGIS image service, Open Data DC. Accessed 17 July 2026. https://imagery.dcgis.dc.gov/dcgis/rest/services/UFD/Urban_Tree_Canopy_2020/ImageServer

[8] District Department of Transportation. 2023. Traffic Volume—2023. ArcGIS feature layer, Open Data DC. Accessed 17 July 2026. https://opendata.dc.gov/datasets/DCGIS::2023-traffic-volume/about

[9] District of Columbia. [n. d.]. Sustainable DC 2.0: Transportation. Historical commute mode share for 2016; accessed 28 October 2026. https://sustainable.dc.gov/sites/default/files/dc/sites/sustainable/page_content/attachments/SDC2%20Transportation.pdf

[10] European Environment Agency. [n. d.]. Thermal Comfort Indices—Universal Thermal Climate Index, 1979–2020. Archived indicator documentation; standard UTCI stress categories; accessed 28 October 2026. https://climate-adapt.eea.europa.eu/en/metadata/indicators/thermal-comfort-indices-universal-thermal-climate-index-1979-2019

[11] A. A. Hagberg, D. A. Schult, and P. J. Swart. 2008. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference*. 11–15. doi:10.25080/TCWV9851

[12] H. G. Kamath, N. Sudharsan, M. Singh, N. Wallenberg, F. Lindberg, and D. Niyogi. 2026. SOLWEIG-GPU: GPU-Accelerated Thermal Comfort Modeling Framework for Urban Digital Twins. *Journal of Open Source Software* 11, 118 (2026), 9535. doi:10.21105/joss.09535

[13] N. Kolaxidis, C. Ludwig, S. Knoblauch, J. Fürle, K. Foshag, S. Fendrich, S. Lautenbach, and A. Zipf. 2025. Mitigating heat stress by reducing solar exposure in pedestrian routing. *Transactions in GIS* 29, 6 (2025), e70110. doi:10.1111/tgis.70110

[14] M. Levenson, D. Pearlmutter, and O. Aleksandrowicz. 2025. An observational analysis of shade-related pedestrian activity. *Buildings and Cities* 6, 1 (2025), 398–414. doi:10.5334/bc.574

[15] F. Lindberg, B. Holmer, and S. Thorsson. 2008. SOLWEIG 1.0: Modelling spatial variations of 3D radiant fluxes and mean radiant temperature in complex urban settings. *International Journal of Biometeorology* 52, 7 (2008), 697–713. doi:10.1007/s00484-008-0162-7

[16] NOAA Office for Coastal Management. 2022. 2022 DC OCTO Lidar DEM: Washington, DC. NOAA InPort item 69939. Accessed 17 July 2026. https://www.fisheries.noaa.gov/inport/item/69939

[17] NOAA Office for Coastal Management. 2022. 2022 DC OCTO Lidar: Washington, DC. NOAA InPort item 69938. Accessed 17 July 2026. https://www.fisheries.noaa.gov/inport/item/69938

[18] OpenStreetMap contributors. 2026. OpenStreetMap. Open database licensed under ODbL 1.0. Accessed 17 July 2026. https://www.openstreetmap.org/copyright

[19] A. Patel, N. Sudharsan, T. Brooks, H. Kamath, D. Crawley, Z. Baumer, M. Coudert, and D. Niyogi. 2026. Thermal comfort path planning tool for urban mobility in Austin, Texas. arXiv preprint arXiv:2602.02540. arXiv:2602.02540 doi:10.48550/arXiv.2602.02540

[20] Georgios Skoumas, Klaus Arthur Schmid, Gregor Jossé, Matthias Schubert, Mario A. Nascimento, Andreas Züfle, Matthias Renz, and Dieter Pfoser. 2015. Knowledge-Enriched Route Computation. In *Advances in Spatial and Temporal Databases (Lecture Notes in Computer Science, Vol. 9239)*. Springer, Cham, 157–176. doi:10.1007/978-3-319-22363-6_9

[21] A. Vartholomaios. 2023. Follow the shade: Detection of optimally shaded pedestrian paths within the historic city center of Thessaloniki. *IOP Conference Series: Earth and Environmental Science* 1196 (2023), 012070. doi:10.1088/1755-1315/1196/1/012070

[22] D. Verma, O. Mumm, and V. M. Carlow. 2026. CoolPaths: Street-scale Physiological Equivalent Temperature (PET) mapping and cooler-routes planning using open data. *City and Environment Interactions* 30 (2026), 100349. doi:10.1016/j.cacint.2026.100349

[23] H. Wolf, A. R. Vierø, and M. Szell. 2025. CoolWalks for active mobility in urban street networks. *Scientific Reports* 15 (2025), 14911. doi:10.1038/s41598-025-97200-2

[24] L. Yang, T. Miyata, S. Kuroki, T. Urakubo, and P. Finnerty. 2025. Route planning in urban environments: A shade-aware navigation framework. In *2025 IEEE 14th Global Conference on Consumer Electronics (GCCE)*. 644–645. doi:10.1109/GCCE65946.2025.11275334

[25] P. Zippenfenig. 2023. Open-Meteo.com Weather API. Computer software, Zenodo. doi:10.5281/zenodo.7970649

A Engineering Feature Definitions

Let $\mathrm{clip}(x) = \min(1, \max(0, x))$. Canopy and land cover are sampled on the densified centerline; building height is additionally sampled at lateral offsets of one-half and one estimated street width.

Let c_e be the fraction of canopy samples at least 2 m high; k_e the mean canopy height divided by 12 m and clipped; v_e the larger of canopy- and land-cover-derived greenery fractions; and w_e the water fraction. The scenic proxy is

$$g_e = \mathrm{clip}(0.60c_e + 0.25k_e + 0.10v_e + 0.05w_e). \tag{9}$$

For nearest-midpoint AADT a_e and the traffic layer's maximum $a_{\max}$, the traffic proxy is $n_e = \log_{10}(a_e + 1)/\log_{10}(a_{\max} + 1)$. This is not a decibel scale.

For solar altitude α, azimuth θ, bearing β_e, and estimated street width W_e, define

$$q_e(z) = \mathrm{clip}\left(\frac{z|\sin(\theta - \beta_e)|}{W_e \max(\tan\alpha, 10^{-6})}\right). \tag{10}$$

If explicit width is missing, $W_e = \max(4, 3.2p_e)$ m for p_e lanes, or 10 m if lane count is also missing. With mean and maximum canopy/building heights, use $z_e^c = \max(\bar{z}_e^c, 0.5z_e^{c,\max})$ and $z_e^b = \max(\bar{z}_e^b, 0.6z_e^{b,\max})$. The geometry shade proxy is

$$s_e^{\mathrm{geom}} = \mathrm{clip}\Big(\max\{c_e, q_e(z_e^b), c_e q_e(z_e^c)\}\Big), \tag{11}$$

set to one below the horizon. For local fractional hour t,

$$\tau_c(t) = [25.5 + 3.2\cos(2\pi(t - 14)/24)]^\circ\mathrm{C}, \tag{12}$$

$$s_e^{\mathrm{cool}} = (n_e^u)^{-1} \sum_i \mathbf{1}[u_{ei} \le \tau_c(t)], \tag{13}$$

$$s_e^{\mathrm{az}} = \max(s_e^{\mathrm{cool}}, s_e^{\mathrm{geom}}). \tag{14}$$

At $t = 14$, the cool threshold is 28.7°C. These coefficients define a hybrid surrogate, not a fitted shadow model.

B Composite Cost

Let signed elevation change be Δz_e, grade $m_e = \Delta z_e/l_e$, and $r_e = 10|m_e| + [\Delta z_e]_+/(25\,\mathrm{m})$. Edge time is t_e, estimated as $l_e/(1.35\,\mathrm{m\,s^{-1}})$ when unavailable. With ϕ_e from Eq. 7, the default cost is

$$A_e = 0.15\frac{l_e}{1000\,\mathrm{m}} + 0.20\frac{t_e}{600\,\mathrm{s}}, \tag{15}$$

$$C_e = A_e + \frac{\max(l_e, 1\,\mathrm{m})}{100\,\mathrm{m}}\big[\phi_e + 0.45(1 - g_e)^2 + 0.45n_e^2 + 0.35r_e + 0.15\sigma_e\big], \tag{16}$$

$$C(P) = \sum_{e \in P} C_e. \tag{17}$$

All ratios are dimensionless. $\sigma_e = 0$ throughout this experiment. Table 3 changes the corresponding coefficients and the shade multiplier. The 1 m floor prevents a zero environmental scale on a malformed zero-length edge. These formulas specify the intended engineering score; Section 3.4 describes the audited projection discrepancy and the uncertainty about the historical execution version.